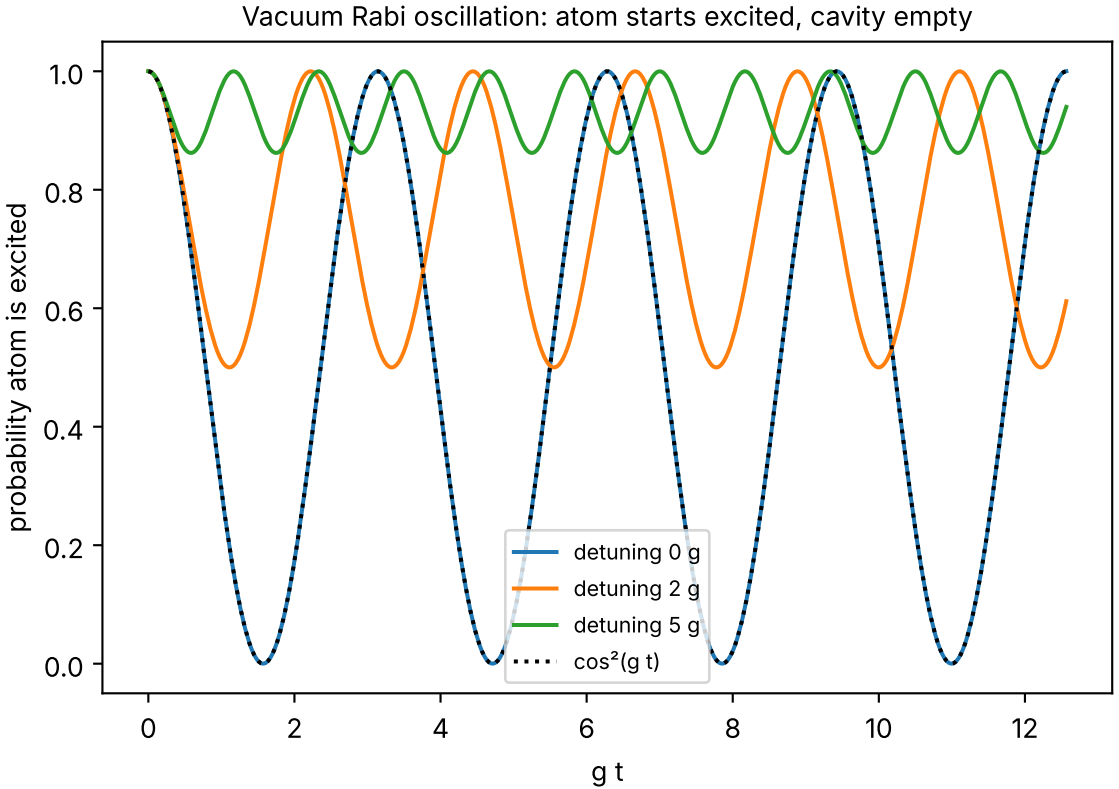
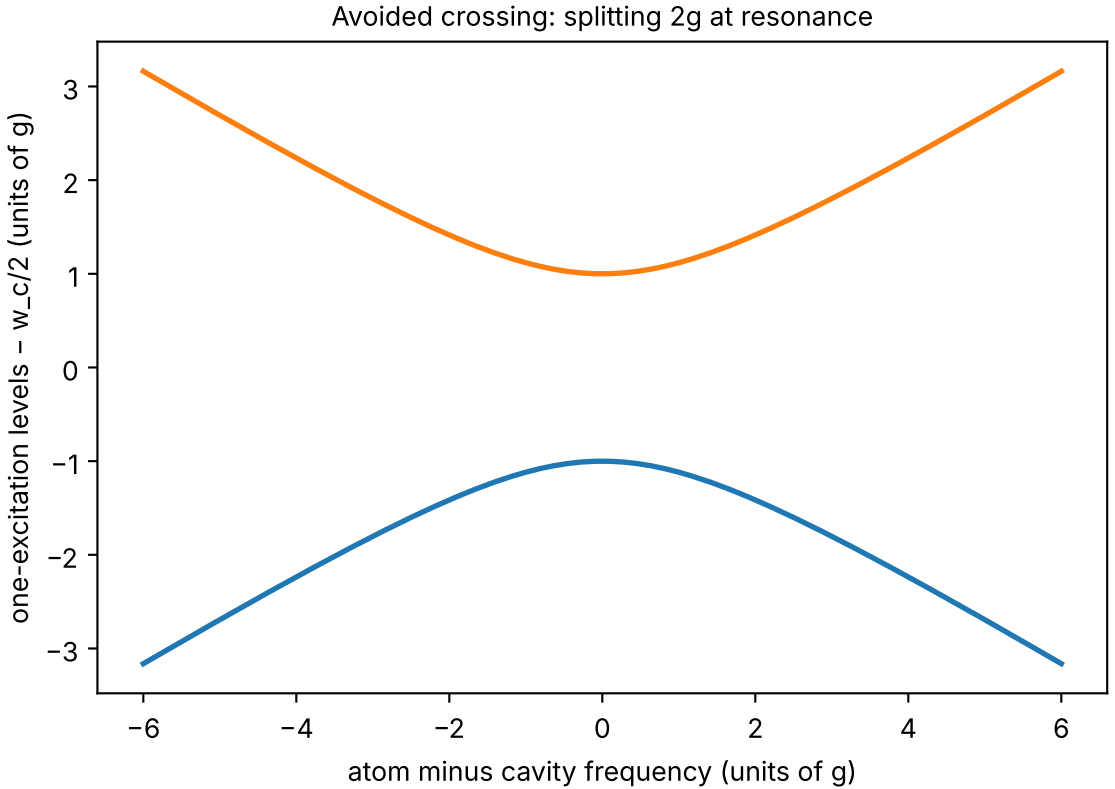

Vacuum Rabi oscillation: atom starts excited, cavity empty
probability atom is excited
g t
detuning 0 g
detuning 2 g
detuning 5 g
cos²(g t)
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
12


Avoided crossing: splitting 2g at resonance
one-excitation levels − w_c/2 (units of g)
atom minus cavity frequency (units of g)
3
2
1
0
−1
−2
−3
−6
−4
−2
0
2
4
6